

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/1

PAPER 1 November 2020

1 hour

PAPER 1 — MULTIPLE CHOICE — 40 QUESTIONS — 1 HOUR

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2020 5006/1.

Additional materials: Answer all 40 questions. For each question there are four possible answers A, B, C and D.

INSTRUCTIONS: Choose the one you consider correct and record it. Each question carries 1 mark.

1. The SI unit of energy is the [1]

- A) joule
- B) newton
- C) watt
- D) pascal

2. An enzyme is [1]

- A) a carbohydrate that stores energy
- B) a biological catalyst
- C) a type of mineral ion
- D) a waste product of respiration

3. Which change increases the rate of photosynthesis, other factors being equal? [1]

- A) decreasing light intensity
- B) placing the plant in darkness
- C) removing chlorophyll
- D) increasing carbon dioxide (up to a point)

4. The SI unit of speed is [1]

- A) kg
- B) N
- C) m/s
- D) J

5. Which statement is correct? [1]

- A) Conduction occurs only in gases
- B) Convection occurs in solids only
- C) Radiation does not need a medium
- D) Conduction is bulk movement of a fluid

6. Cholera is spread mainly by [1]

- A) contaminated water / food
- B) sexual contact only
- C) air droplets only
- D) mosquitoes

7. The kidney [1]

- A) stores glycogen only
- B) makes red cells
- C) produces bile
- D) removes urea and excess water from the blood

8. The pitch of a sound depends on its [1]

- A) frequency
- B) colour
- C) amplitude
- D) loudness only

9. Most carbon dioxide for photosynthesis enters the leaf through the [1]

- A) xylem
- B) root hairs
- C) cuticle
- D) stomata

10. A metal kettle is safer if it has [1]

- A) an earth wire and a fuse / circuit breaker
- B) a higher current always
- C) a thicker live wire only
- D) no insulation

11. Red blood cells are adapted to carry oxygen because they contain [1]

- A) starch
- B) chlorophyll
- C) haemoglobin
- D) cellulose

12. The law of reflection states that [1]

- A) light always bends towards the normal in air
- B) angle of incidence = angle of reflection
- C) angle of incidence = angle of refraction always
- D) a mirror absorbs all light

13. A unit of density is [1]

- A) W
- B) J
- C) N
- D) g/cm³

14. Electrolysis requires [1]

- A) a solid ionic compound with no mobile ions
- B) pure water with no ions, always
- C) a sugar solution
- D) a molten or aqueous ionic compound

15. Convection occurs in [1]

- A) fluids (liquids and gases)
- B) crystals only
- C) a vacuum
- D) solids only

16. Carbon dioxide in air is tested with **[1]**

- A) limewater, which turns milky
- B) blue litmus only
- C) iodine
- D) a lighted splint

17. During electrolysis, cations move to the **[1]**

- A) voltmeter
- B) cathode
- C) fuse
- D) anode

18. Convection occurs in **[1]**

- A) solids only
- B) a vacuum
- C) crystals only
- D) fluids (liquids and gases)

19. The halogens are in **[1]**

- A) Group 0
- B) Group 7 / 17
- C) Period 1 only
- D) Group 1

20. A learner runs 20 m in 5 s. Average speed is **[1]**

- A) 5 m/s
- B) 25 m/s
- C) 4 m/s
- D) 100 m/s

21. Hydrochloric acid in the stomach **[1]**

- A) digests cellulose
- B) stores bile
- C) neutralises bile
- D) kills many bacteria and provides a low pH for protease

22. Which is correct? **[1]**

- A) silver displaces sodium
- B) magnesium displaces copper from $\text{CuSO}_4(\text{aq})$
- C) copper displaces magnesium from $\text{MgSO}_4(\text{aq})$
- D) gold displaces zinc

23. A metal kettle is safer if it has **[1]**

- A) a higher current always
- B) a thicker live wire only
- C) an earth wire and a fuse / circuit breaker
- D) no insulation

24. The period number of an element is the number of **[1]**

- A) protons only
- B) outer electrons
- C) neutrons
- D) occupied electron shells

25. Distilled water, compared with tap water, contains **[1]**

- A) sand
- B) more chlorine always
- C) fewer dissolved salts
- D) more dissolved salts

26. Diffusion is fastest in a **[1]**

- A) crystal lattice at 0 K
- B) liquid
- C) gas
- D) solid

27. In double circulation, blood passes through the heart **[1]**

- A) once per circuit of the body
- B) twice per circuit of the body
- C) only at night
- D) not at all

28. A mass of 4 kg accelerates at 3 m/s^2 . The unbalanced force is **[1]**

- A) 0 N
- B) 7 N
- C) 40 N
- D) 12 N

29. The function of the nucleus is to **[1]**

- A) be a fluid-filled sac only
- B) make glucose
- C) control cell activities
- D) store starch only

30. The main function of red blood cells is to **[1]**

- A) transport oxygen
- B) fight all pathogens by phagocytosis only
- C) clot blood
- D) make urea

31. A lighted splint is used to test for hydrogen. A positive result is **[1]**

- A) a pop sound
- B) limewater turning milky
- C) bleaching litmus
- D) a glowing splint relighting

32. When a metal is oxidised, **[1]**

- A) the metal always melts
- B) pH becomes 7 always
- C) oxygen is added / electrons are lost
- D) oxygen is removed

33. In human muscle during vigorous exercise, anaerobic respiration produces **[1]**

- A) oxygen
- B) ethanol only
- C) carbon dioxide and water
- D) lactic acid

34. The gravitational force on an object is its **[1]**

- A) volume
- B) mass
- C) weight
- D) density

35. A plant cell in a concentrated sugar solution may become **[1]**

- A) plasmolysed
- B) turgid
- C) photosynthetic
- D) a gamete

36. The electronic configuration of magnesium (proton number 12) is **[1]**

- A) 2,8,1
- B) 2,8,7
- C) 2,8,2
- D) 2,6

37. Pollination is **[1]**

- A) germination of the seed
- B) transfer of pollen from anther to stigma
- C) dispersal of seeds
- D) fusion of gametes

38. Which organelle is the main site of aerobic respiration? **[1]**

- A) mitochondrion
- B) cell wall
- C) vacuole
- D) chloroplast

39. Universal indicator in a strong alkali is **[1]**

- A) purple / blue
- B) colourless always
- C) red
- D) green

40. Universal indicator in a strong alkali is **[1]**

- A) purple / blue
- B) green
- C) red
- D) colourless always

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).